

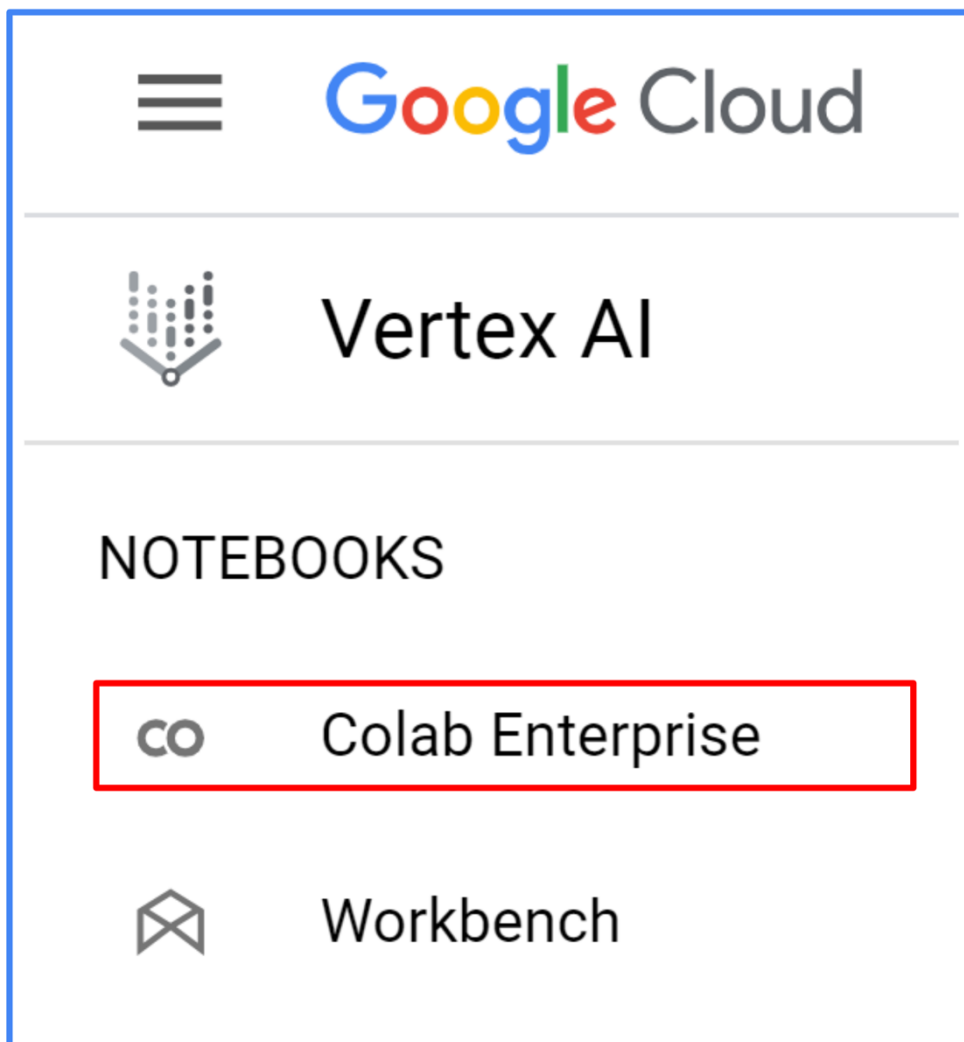
Task 1. Set up your environment

1. Enable the Vertex AI API

Navigate to the [Vertex AI section of your Cloud Console](#) and click **ENABLE ALL RECOMMENDED APIS**.

Task 2. Create a Colab Enterprise Notebook

1. In the Vertex AI section, scroll down to Notebooks. Click **Colab Enterprise**.



A "Welcome to Colab Enterprise" page then appears.

Welcome to Colab Enterprise

Quick actions

 Create notebook

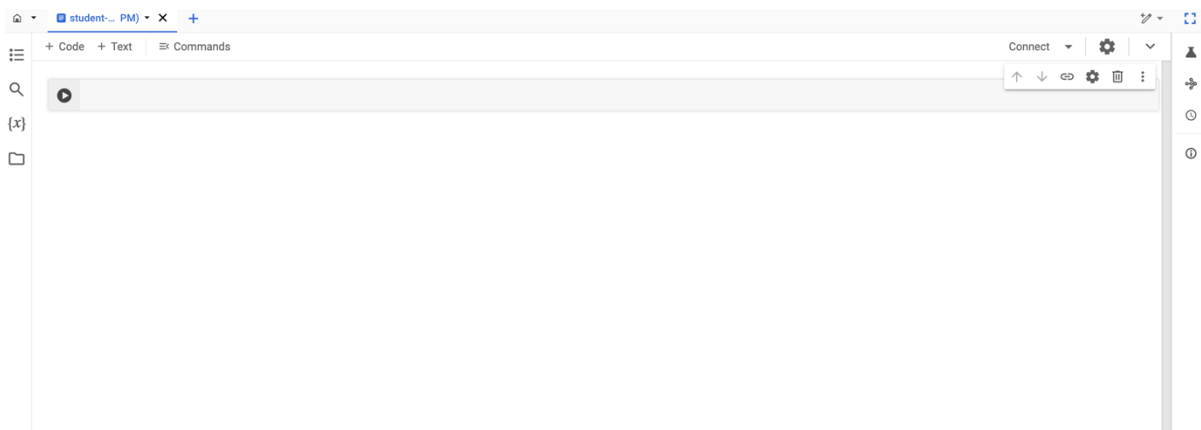
 Import notebooks from Cloud Storage

 Import notebooks from URLs

 Upload notebooks

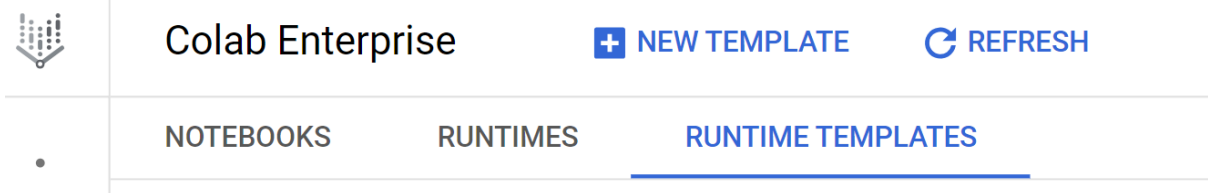
2. In the **Region** menu, select the region **us-central1**, where you create the notebook.
3. Click **+ CREATE NOTEBOOK** in Quick actions section.

A new notebook will opened.



In order to execute the cells you need to create a runtime. Recall that runtimes are instances derived from runtime templates that allow users to run Colab notebooks. To create a runtime, you need to first create a runtime template.

4. Click **RUNTIME TEMPLATES**. On the Runtime Templates page, click **+ NEW TEMPLATE**.



Runtime Basics

There are three steps. Step 2 and Step 3 are optional.

Step 1: Provide runtime basic information and select region as **us-central1**

← Create new runtime template

- **Runtime basics**
Begin by providing basic information
- **Configure compute**
Set machine, accelerator and disk properties
- **Networking and security**
Specify network and security settings

Display name *

i Some regions are restricted due to a policy set by your organization.

Region *
us-central1 (Iowa)

Description

Optional

Labels

Optionally add labels to identify and group Runtime Templates

+ ADD LABEL

CONTINUE

Compute Configure

Step 2: Configure Compute (Optional)

← Create new runtime template

- ✓ **Runtime basics**
Begin by providing basic information
- **Configure compute**
Set machine, accelerator and disk properties
- **Networking and security**
Specify network and security settings

Choose machine configuration

Machine type *

e2-standard-4

Disk type *

Standard Disk (pd-standard)

Disk size *

100

GB

☐ Secure boot **?**

Verifies the digital signature of all boot components to protect against boot-level and kernel-level malware. Secure boot runtimes cannot be upgraded.

Encryption

☒ Google-managed encryption key

Keys owned by Google

☐ Cloud KMS key

Keys owned by customers

Idle shutdown

☒ Enable Idle Shutdown **?**

Time of inactivity before shutdown (Minutes) *

180

Range: 10 minutes to 1 day (1,440 minutes)

BACK

CONTINUE

Networking and Security

Step 3: Networking and Security (Optional)

← Create new runtime template

☒ Runtime basics
Begin by providing basic information

☒ Configure compute
Set machine, accelerator and disk properties

☐ **Networking and security**
Specify network and security settings

Choose networking configuration

Network ▼

Subnetwork ▼

☒ Enable public internet access ?

☒ Enable end-user credentials

Network tags

BACK

Click **Create**.

Task 3. Run Code in a Colab Enterprise Notebook

In this example, the code cell below “Getting Started” uses numpy to generate some random data and uses matplotlib to visualize it.

1. Click **Colab Enterprise** then click on your created Notebook.
2. Copy the code below and click **Run** icon to run the cell.

```
3.     import numpy as np
4.     from matplotlib import pyplot as plt
5.
6.     ys = 200 + np.random.randn(100)
7.     x = [x for x in range(len(ys))]
8.
```

```
9.     plt.plot(x, ys, '-')
10.     plt.fill_between(x, ys, 195, where=(ys > 195),
    facecolor='g', alpha=0.6)
11.
12.     plt.title("Sample Visualization")
    plt.show()
```

Copied!



```
import numpy as np
from matplotlib import pyplot as plt

ys = 200 + np.random.randn(100)
x = [x for x in range(len(ys))]

plt.plot(x, ys, '-')
plt.fill_between(x, ys, 195, where=(ys > 195), facecolor='g', alpha=0.6)

plt.title("Sample Visualization")
plt.show()
```

As you execute the cell, a message pops up indicating that the runtime is active and initiating a connection - as shown in the image below.

Runtime is active, initiating connection... Colab Enterprise is authorized with your Google account and has access to Google Cloud Services. Review the code [Dismiss](#) before executing.

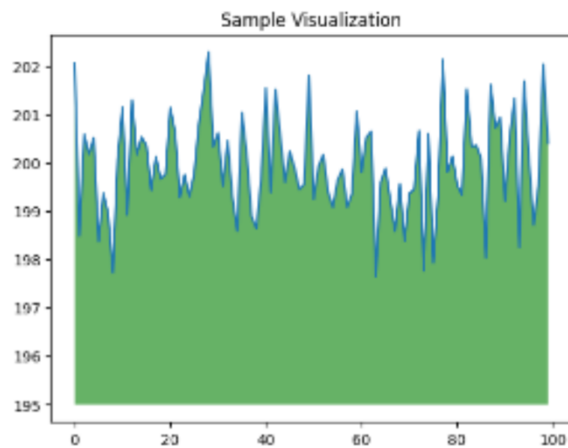
3. Click **Open** for **Open OAuth popup** then select your **User Name** then click **Allow**.
4. Now check the cell you executed, there should be a green check mark next to, this indicates that the cell executed properly.

```
import numpy as np
from matplotlib import pyplot as plt

ys = 200 + np.random.randn(100)
x = [x for x in range(len(ys))]

plt.plot(x, ys, '-')
plt.fill_between(x, ys, 195, where=(ys > 195), facecolor='g', alpha=0.6)

plt.title("Sample Visualization")
plt.show()
```



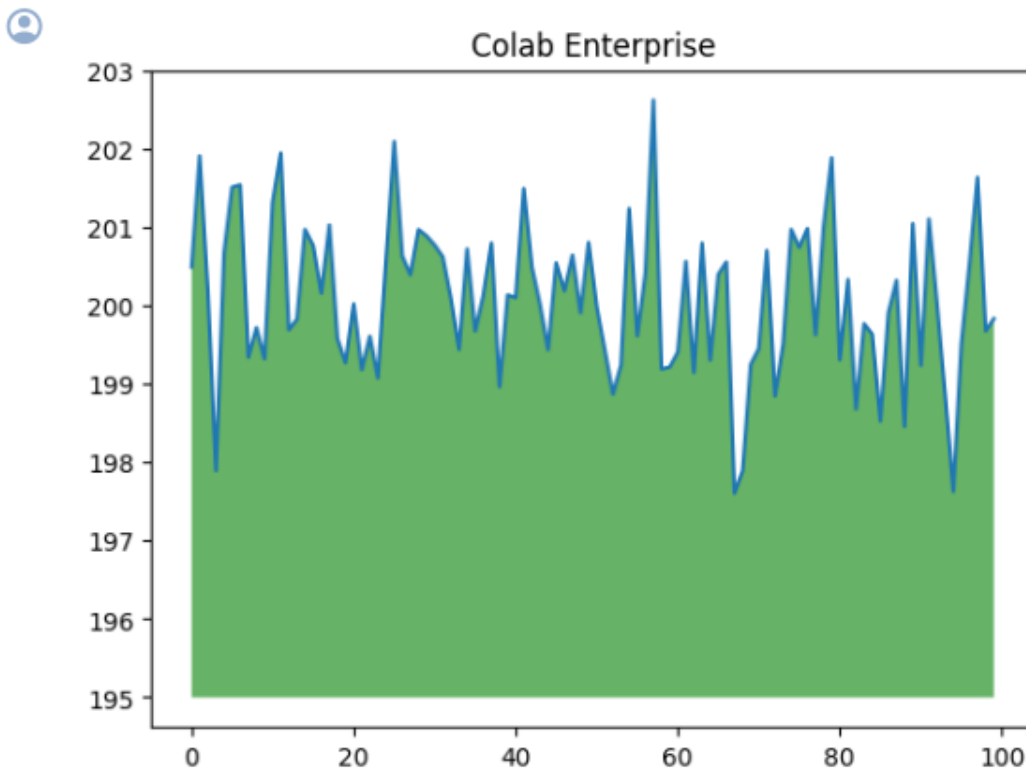
5. Now, make a change to the code - for example, change the title of the plot from "Sample Visualization" to "Colab Enterprise". Then, execute the cell.

```
import numpy as np
from matplotlib import pyplot as plt

ys = 200 + np.random.randn(100)
x = [x for x in range(len(ys))]

plt.plot(x, ys, '-')
plt.fill_between(x, ys, 195, where=(ys > 195), facecolor='g', alpha=0.6)

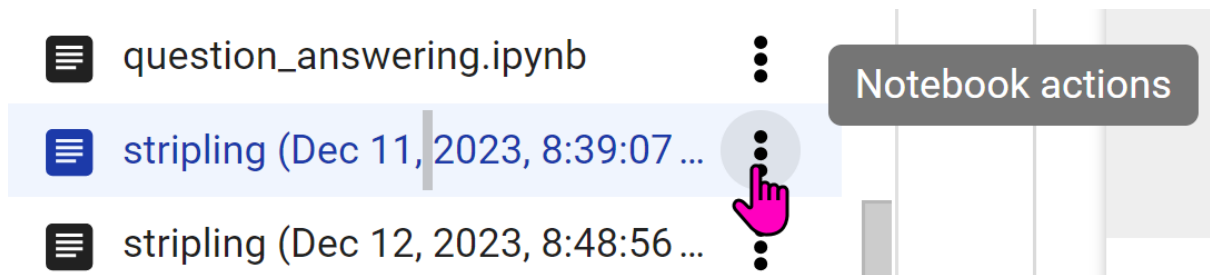
plt.title("Colab Enterprise")
plt.show()
```



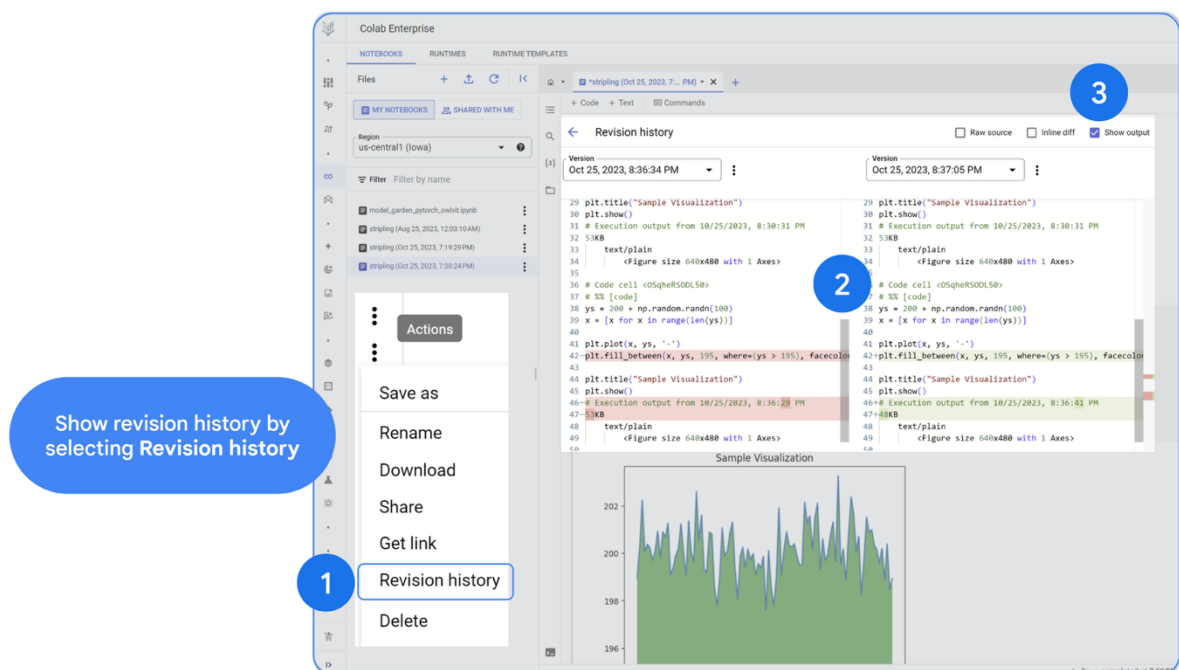
Task 4. Show revision history

One of the most important features of software development is the ability to track version history.

1. Go to the Notebook Storage section. Click on the **three dots** beside the notebook you created.



2. Select **Revision History** (as shown in bubble #1 in the image below).
 When revision history is selected, you see the changes side by side with a date stamp and color-coding to see the “old” in red and the “new” in green (as shown in bubble #2).
 There are three options to view revision history: the raw source, the inline differences, or the source output (as shown in bubble #3).



Task 5. Add code to cells

To add code or text to a Notebook, simply click on either code or text in the menu bar above the Notebook Editor.



Now, you will add several blocks of code to the notebook. After you copy a block of code, run the code to see the output. Note - Some cells will have not output (such as when you import the libraries). When you are done, share the notebook.

1. Go back to the **Notebook storage section** then click on your Notebook.
2. Re-run the first code again.
3. Add a code cell.
4. Copy the code below into the new cell.

Import libraries

```
import seaborn as sns
import pandas as pd
import numpy as np
from google.cloud import bigquery
bq = bigquery.Client()
Copied!
```

5. Click the **Run** icon to run the cell.
- There should be no output shown.

Insert the following code as cells to import the necessary modules and initialize a BigQuery client. The BigQuery client will be used to send and receive messages from the BigQuery API.

Import BigQuery Client

```
client = bigquery.Client()

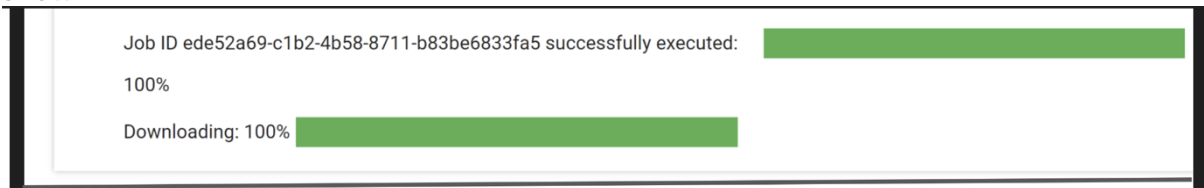
query = """SELECT * FROM `bigquery-public-
data.catalonian_mobile_coverage_eu.mobile_data_2015_2017` LIMIT 1000"""
job = client.query(query)
df = job.to_dataframe()
Copied!
No output shown.
```

Download a BigQuery Table into a Pandas Dataframe

In Google, %%bigquery is a magic command used within Jupyter notebooks and other interactive environments to interact with BigQuery. Therefore, %%bigquery essentially tells your environment to switch to BigQuery mode: It prepares the environment to accept and execute BigQuery queries. BigQuery will execute the query, retrieve the data, and present it to you within the notebook environment, often as a pandas DataFrame (e.g. the "df" as shown in the code)

```
%%bigquery df
SELECT *
FROM `bigquery-public-
data.catalonian_mobile_coverage_eu.mobile_data_2015_2017`
```

Copied!
Output
shown:



Show the first five rows of the Pandas Dataframe

`df.head()`
Copied!
Output
shown:

✓ 0s `df.head()`

	date	hour	lat	long	signal	network	operator	status	description	net	speed	s
0	2015-01-18	12:49:17	39.23566	-0.41185	12	null	null	2	STATE_EMERGENCY_ONLY	3G	10.2	
1	2015-01-18	17:46:02	41.53350	2.15793	14	null	null	2	STATE_EMERGENCY_ONLY	3G	0.0	
2	2015-01-18	17:56:44	41.40443	2.16549	20	null	null	2	STATE_EMERGENCY_ONLY	3G	3.7	
3	2017-10-09	10:07:43	41.87279	1.87672	14	Sin servicio	Sin servicio	2	STATE_EMERGENCY_ONLY	3G	115.9	
4	2015-02-04	18:50:38	41.43143	2.19309	26	null	null	2	STATE_EMERGENCY_ONLY	2G	3.4	

Get information on the Pandas Dataframe

`df.info()`
Copied!

Output
shown:

✓
0s



df.info()



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11744914 entries, 0 to 11744913
Data columns (total 20 columns):
#   Column                Dtype
---  -
0   date                  dbdate
1   hour                  dbtime
2   lat                   float64
3   long                  float64
4   signal                Int64
5   network               object
6   operator              object
7   status                Int64
8   description            object
9   net                   object
10  speed                 float64
11  satellites             float64
12  precission            float64
13  provider              object
14  activity               object
15  downloadSpeed          float64
16  uploadSpeed            float64
17  postal_code            object
```

Get statistics on the Pandas Dataframe

df.describe()
Copied!

Output
shown:

```
df.describe()
```

	lat	long	signal	status	speed	satellites	precision	downloadSpeed	uploadSpeed
count	1.174491e+07	1.174491e+07	11744914.0	11744914.0	1.174491e+07	1.174369e+07	1.174369e+07	0.0	0.0
mean	4.120760e+01	7.851310e-01	13.262954	1.778794	2.643376e+01	9.795352e+05	4.636630e+07	NaN	NaN
std	3.481886e+00	9.397493e+00	7.467304	0.633283	3.553885e+01	3.356759e+09	9.644288e+10	NaN	NaN
min	-7.703775e+01	-1.787292e+02	0.0	0.0	0.000000e+00	0.000000e+00	0.000000e+00	NaN	NaN
25%	4.138028e+01	1.690790e+00	8.0	2.0	1.400000e+00	2.000000e+00	1.000000e+01	NaN	NaN
50%	4.148911e+01	2.123840e+00	13.0	2.0	7.400000e+00	5.000000e+00	1.700000e+01	NaN	NaN
75%	4.169777e+01	2.245140e+00	18.0	2.0	4.080000e+01	7.000000e+00	3.000000e+01	NaN	NaN
max	6.343513e+01	1.396410e+02	99.0	3.0	2.550000e+02	1.150330e+13	2.415039e+14	NaN	NaN

Plot a correlation using Seaborn.

```
numeric_df = df.select_dtypes(include=[np.number])
```

```
corr_matrix = numeric_df.corr()
```

```
plt.figure(figsize=(10, 5))
```

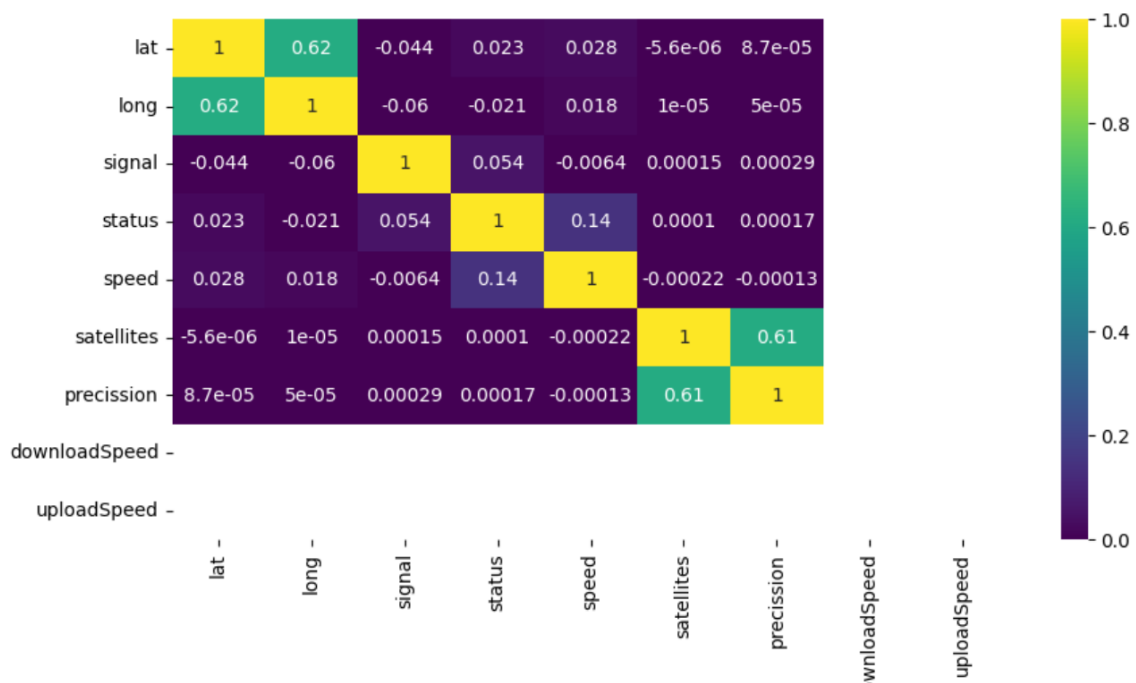
```
sns.heatmap(corr_matrix, annot=True, vmin=0, vmax=1, cmap='viridis')
```

```
plt.show()
```

Copied!

Output

shown:



Write a SQL query to pick up specific fields from a BigQuery dataset

```
%%bigquery df2
```

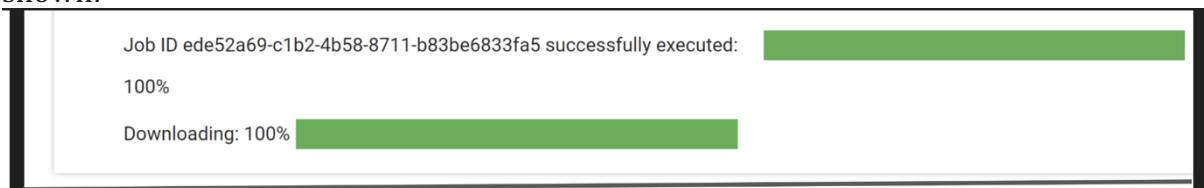
```
SELECT signal, status
```

```
FROM `bigquery-public-  
data.catalonian_mobile_coverage_eu.mobile_data_2015_2017`
```

Copied!

Output

shown:



Get the first five rows of the new fields.

```
df2.head()
```

Copied!

Output

shown:

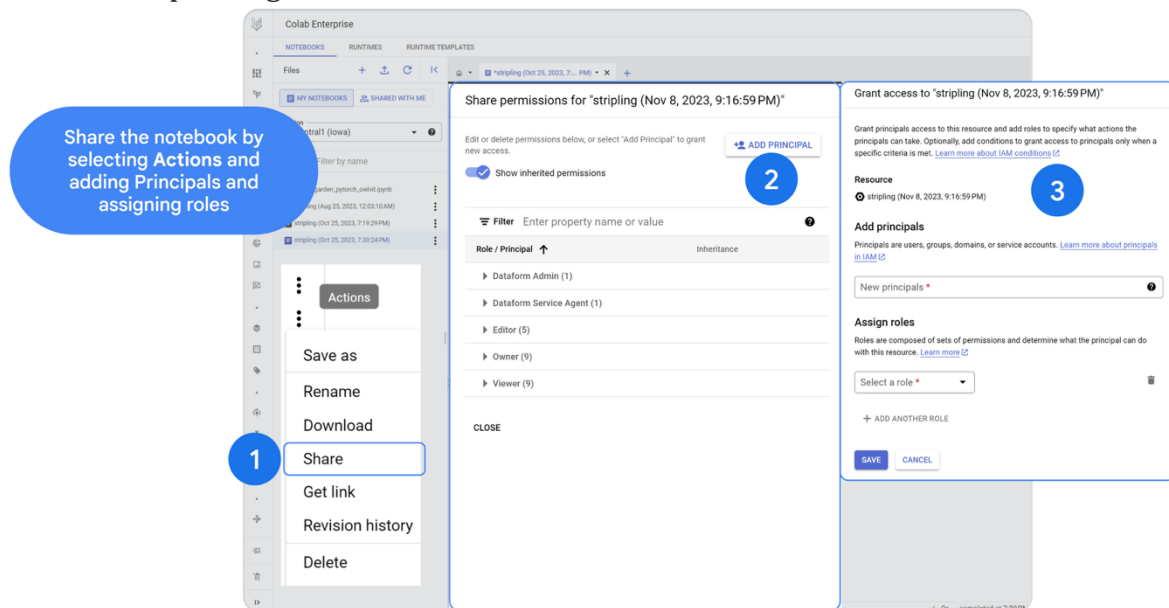
	signal	status
0	52	2
1	58	0
2	53	0
3	51	0
4	56	0



Task 6. Share the Notebook

Next, share your notebook.

1. When you select “Share” by right-clicking on the notebook, a share permissions window appears, which allows you to edit or delete permissions, or select "Add Principal" to grant new access.



2. When you grant principals access to a resource, you also add roles to specify what actions the principals can take. Optionally, you can add conditions to grant access to principals only when a specific criteria is met.

Principals are users, groups, domains, or service accounts. Roles are composed of sets of permissions and determine what the principal can do with this resource.

Congratulations!

In this lab you learned how to:

- Create a Colab Enterprise Notebook
- Connect to BigQuery datasets
- Perform statistical analysis on a Pandas Dataframe
- Create Seaborn plots for Exploratory Data Analysis in Python
- Write a SQL query to pick up specific fields from a BigQuery dataset
- Use version history to see code changes
- Share a Colab Enterprise notebook

End your lab

When you have completed your lab, click **End Lab**. Google Skills removes the resources you've used and cleans the account for you.

You will be given an opportunity to rate the lab experience. Select the applicable number of stars, type a comment, and then click **Submit**.

The number of stars indicates the following:

- 1 star = Very dissatisfied
- 2 stars = Dissatisfied
- 3 stars = Neutral
- 4 stars = Satisfied
- 5 stars = Very satisfied

You can close the dialog box if you don't want to provide feedback.

For feedback, suggestions, or corrections, please use the **Support** tab.

Manual Last Updated December 11, 2024

Lab Last Tested December 11, 2024

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